

SCHOOL SCIENCE PROJECT

NOISE METER

Real-Time Sound Level Monitoring & Noise Alert System



"Every decibel counted, every ear protected."

Student Name · School Name
Science & Innovation Exhibition

Project Domain
Electronics · IoT · Environmental Monitoring

THE PROBLEM

Why unmeasured, unmanaged noise quietly damages health and focus



Rising Urban Noise

Traffic, construction and industry push city sound levels well past safe daily limits.



Health Impact

Prolonged exposure causes hearing loss, stress, poor sleep and high blood pressure.



No Visibility

Most homes, schools and offices have no way to know when noise has crossed a safe limit.



Classroom Disruption

Noisy classrooms and labs reduce concentration and lower learning outcomes for students.

The WHO identifies environmental noise as one of the top environmental risks to public health, with continuous urban exposure above 55 dB linked to sleep disturbance and cardiovascular stress.

PROJECT OBJECTIVES

What the Noise Meter aims to achieve



Measure Sound Levels

Continuously capture ambient noise and convert it to an approximate decibel (dB) reading.



Detect Threshold Breach

Compare live readings against safe-limit thresholds for homes, schools and workplaces.



Instant Visual Alert

Drive a colour-coded LED / display so anyone can see noise status at a glance.



Audible Alarm

Trigger a buzzer the moment noise crosses an unsafe level, prompting quick action.



Log & Trend Data

Record readings over time to reveal noisy hours, zones or repeated violations.



Low-Cost & Scalable

Build with affordable, easily available components suitable for schools and homes.

PROPOSED SOLUTION

A compact device that listens, measures, and warns — before noise causes harm

Concept Overview

- A sound sensor module (microphone + amplifier) continuously picks up ambient noise.
- A microcontroller (Arduino / ESP32) samples the signal and estimates the sound level in decibels.
- Readings are shown live on an OLED/LCD display and mapped to a Green–Yellow–Red status.
- When noise exceeds a safe threshold, an LED and buzzer trigger an instant alert.
- Optional WiFi/Bluetooth module streams readings to a mobile app or dashboard for logging.

Operating Loop

1



Sense

Microphone module picks up ambient sound continuously

2



Process

Microcontroller converts signal into an estimated dB value

3



Display

Live reading & colour status shown on OLED / LED strip

4



Alert / Log

Buzzer sounds on breach; data optionally logged / sent out

SYSTEM ARCHITECTURE

Core hardware modules inside the Noise Meter unit



Sound Sensor Module

Electret microphone + op-amp (e.g. KY-038) captures ambient sound amplitude.



Microcontroller (Arduino/ESP32)

Central brain — samples audio, computes dB estimate, drives outputs.



OLED / LCD Display

Shows live sound level and Green-Yellow-Red safety status.



Buzzer + RGB LED

Audible + visual alarm the instant a threshold is crossed.



WiFi / Bluetooth Module

Streams live readings to a phone app or dashboard for logging.



Rechargeable Battery Pack

Portable power supply with voltage regulation for all modules.

KEY FEATURES

What sets this Noise Meter apart



Real-Time dB Reading

Continuous sound-level sampling with near-instant display updates.



Smart Threshold Alerts

Configurable safe limits for home, classroom or industrial settings.



Data Logging & Trends

Captures noise history to reveal peak-noise hours and patterns.



Remote Monitoring

Optional WiFi link streams readings to a dashboard or mobile app.

Design Highlights

30–130

dB range measured

<1s

Alert response time

3

Alert levels (Safe/Caution/Danger)

Low

Component cost — school-lab friendly

APPLICATIONS

Where the Noise Meter can be deployed



Schools & Libraries

Keeps classrooms and study halls within a focus-friendly noise range.



Hospitals & Clinics

Maintains quiet zones for patient rest and recovery wards.



Industrial Workplaces

Flags hazardous noise levels near heavy machinery for worker safety.



Smart City / Traffic

Monitors street-level noise to support urban noise-pollution mapping.

Broader Impact By making invisible noise visible, this low-cost prototype helps schools, hospitals and workplaces respond to unsafe sound levels before they cause lasting harm — a small device pointing to safer, quieter shared spaces.

FUTURE SCOPE & CONCLUSION

Current Prototype

Real-time dB sensing · Green-Yellow-Red alerts · buzzer
+ LED · OLED display

Next Phase

Mobile app dashboard · cloud data logging ·
configurable multi-zone thresholds

Long-Term Vision

Networked noise-mapping across a campus or
neighbourhood · integration with smart-city sensors



"You cannot fix the noise you cannot see — this meter makes it visible."